

Electronics - Semiconductor Physics

Post Circuits

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1 Fundamentals of Atomic Models

1.1 Hydrogen Energy Levels

[Pg 14]

The energy levels for a hydrogen atom are given by:

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

where n is the **principal quantum number**.

In terms of the Rydberg coefficient R_E (for Hydrogen-like energy levels):

[Pg 15]

$$E_n = -\frac{Z^2}{n^2} R_E$$

where:

- Z = nuclear charge
- $n = 1, 2, 3, \dots$

The Rydberg constant is defined as:

$$R_E = \frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2}$$

In atomic units (where $e^2/4\pi\epsilon_0 = 1$):

$$R_E = \frac{m_e}{2\hbar^2}$$

Excitons Excitons (electron-hole pairs) behave like hydrogen-like atoms but are much **larger** in size.

- The Bohr radius is scaled by ϵ_r/m^* .
- The Exciton binding energy is much smaller than 13.6 eV.
- *Implication:* This explains why excitons can be easily thermally ionized at room temperature in many materials.

2 Band Theory of Solids

2.1 Formation of Bands

[Pg 16]

Discrete energy levels of atoms split into bands because of the **overlap** of atomic orbitals.

- The 'valence' (outermost) atomic orbitals form the **widest energy bands** in solids because of **maximum overlap** between neighbouring atoms' wavefunctions.
- Typically, the free electron wavefunction in solid state physics is that of a **plane wave**:

$$\Psi(x) = e^{ikx}$$

Each value of k corresponds to an energy eigenstate:

- Wavelength: $\lambda = \frac{2\pi}{k}$
- Momentum: $p = \hbar k$
- Energy: $E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$

2.2 Bloch's Theorem

Electrons in a crystal (like a semiconductor lattice) are not exactly free. Bloch's theorem states that their wavefunction takes the form: [Pg 17]

$$\Psi_k(x) = U_k(x) \cdot e^{ikx}$$

$\downarrow$ Periodic potential due to the lattice ($U_k(x+a) = U_k(x)$)
 $\downarrow$ Plane wave component

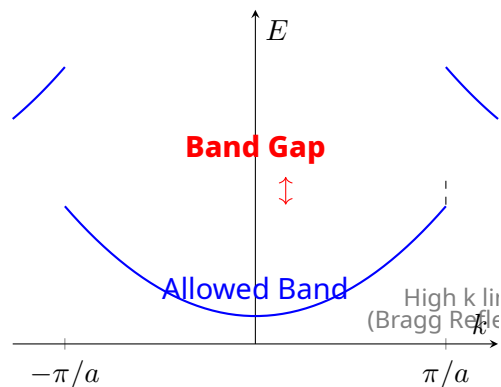
2.2.1 E-k Diagrams: Free vs. Real Solid

1. **Free Electron Model:** Energy increases continuously.

$$E = \frac{\hbar^2 k^2}{2m} \quad (\text{Continuous Parabola})$$

2. **Real Solid (with Periodic Potential):** The parabola splits into bands with **allowed regions** and **forbidden regions**.

Parabola Splitting into Bands



Bandwidth: The range of $E(k)$ within 1 band.

[Pg 18]

2.3 Band Gaps of Some Materials

- **Diamond (Insulator):** $E_g \approx 5.5$ eV
- **Silicon (Si):** $E_g \approx 1.1$ eV
- **GaAs:** $E_g \approx 1.4$ eV

2.3.1 Formation Mechanism

Question: How do Band-gaps form?

Answer: By Bloch's theorem, $\Psi_k(x) = U_k(x)e^{ikx}$.

At the **Brillouin Zone Boundaries** (e.g., $k = \pm\pi/a$ in 1D), the free electron states $e^{i\frac{\pi}{a}x}$ and $e^{-i\frac{\pi}{a}x}$ become **degenerate** in energy.

This results in two degenerate states:

$$|+\rangle = e^{i\frac{\pi}{a}x} \quad \text{and} \quad |-\rangle = e^{-i\frac{\pi}{a}x}$$

[Pg 19]

2.3.2 Effective Hamiltonian Matrix

To lift this degeneracy, we consider the potential coupling V_G . The Hamiltonian matrix is:

$$\hat{H} = \begin{pmatrix} E_0 & V_G \\ V_G^* & E_0 \end{pmatrix}$$

Where:

- $E_0 = \frac{\hbar^2(\pi/a)^2}{2m}$ (Free electron energy at boundary)
- V_G = Off-diagonal entries arising from potential coupling.

Solving for eigenvalues:

$$E^{(1,2)} = E_0 \pm |V_G|$$

Conclusion: The degeneracy is lifted at the Brillouin zone edges, and level splitting occurs.

$$\boxed{\Delta E = 2|V_G|} \leftarrow \text{This is the Band Gap}$$

3 Tight-Binding Model

3.1 Hamiltonian

[Pg 20]

The Tight-Binding Hamiltonian is given by:

$$\hat{H} = \sum_{i,j} t_{ij} c_i^\dagger c_j + V \sum_i n_i$$

Here, t_{ij} represents the **hopping amplitude** (tunneling) between neighbouring atomic sites.

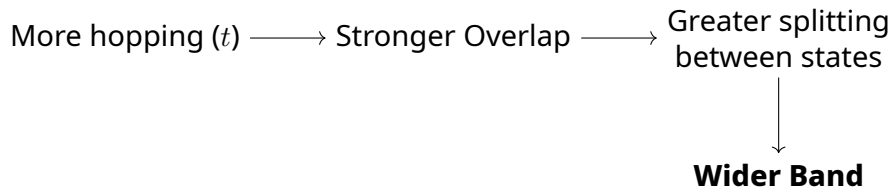
Key Concept: The bandwidth is set by the hopping amplitude t_{ij} .

More hopping $\rightarrow$ More overlap $\rightarrow$ Degeneracy is lifted.

Explanation: Let isolated atoms each have energy E_0 . When we bring them close, their wavefunctions overlap. Hopping (t) mixes the atomic states into linear combinations. Pairwise, degeneracy is lifted.

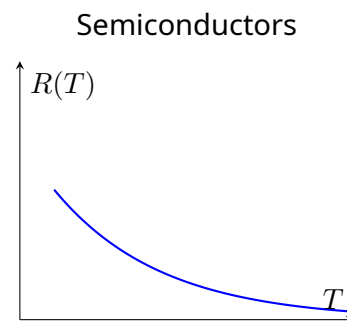
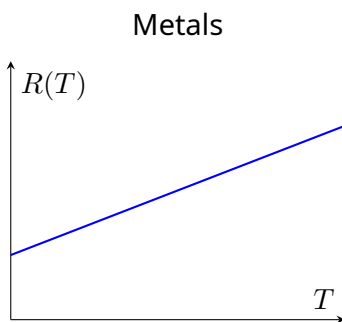
Instead of N -identical energy levels (for N -atoms), we get a **band** of N closely spaced energies.

[Pg 21]



3.2 Temperature Dependence of Resistance

1. **Conductors (Metals):** $R(T) = R_0 + \alpha T + \beta T^3 \dots$
(Resistance increases with Temperature)
2. **Semiconductors:** $R(T) = R_0 e^{E_g/2k_B T}$
(Resistance decreases as Temperature increases)

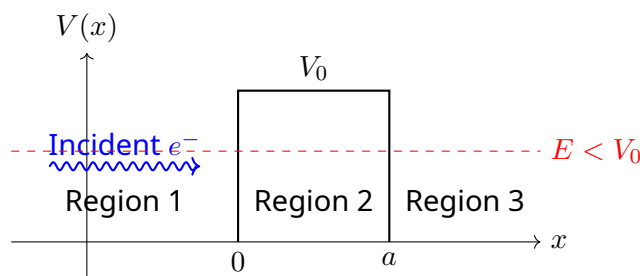


4 Quantum Tunneling

4.1 Problem Setup

[Pg 22]

Consider a potential barrier V_0 of width a .



Wavefunctions in Regions:

- **Region 1 ($x < 0$):** $\Psi_I(x) = Ae^{ikx} + Be^{-ikx}$
(Incident + Reflected)
- **Region 2 ($0 < x < a$):** $\Psi_{II}(x) = Ce^{-\kappa x} + De^{\kappa x}$
(Tunnelling inside barrier)
- **Region 3 ($x > a$):** $\Psi_{III}(x) = Fe^{ikx}$
(Transmitted wave, right-going)

4.2 Schrödinger Equation (TISE)

$$-\frac{\hbar^2}{2m} \frac{d^2\Psi(x)}{dx^2} + V\Psi(x) = E\Psi(x)$$

Rearranging for Region 2 ($V = V_0$):

[Pg 23]

$$-\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} + (V_0 - E)\Psi = 0$$

$$\frac{d^2\Psi}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2} \Psi = 0$$

Let $\kappa^2 = \frac{2m(V_0 - E)}{\hbar^2}$. The solution is:

$$\Psi_{II}(x) = Ce^{-\kappa x} + De^{\kappa x}$$

Typically, for a wide barrier, the decaying term dominates, leading to tunneling.

5 Effective Mass & Carriers

5.1 Definition

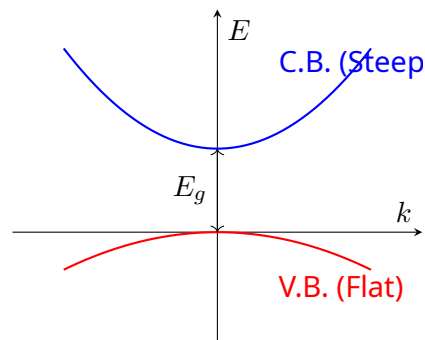
[Pg 24]

We define the **effective mass** m^* related to the curvature of the band:

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \left(\frac{d^2 E}{dk^2} \right)$$

- **High Curvature** (Steep parabola) $\rightarrow$ Large $\frac{d^2 E}{dk^2} \rightarrow$ **Small** m^* (Light carrier).
- **Low Curvature** (Flat parabola) $\rightarrow$ Small $\frac{d^2 E}{dk^2} \rightarrow$ **Large** m^* (Heavy carrier).

5.2 Valence vs. Conduction Bands



- The **Valence Band** is typically flatter than the Conduction Band because holes have a larger effective mass than electrons.
- **Valence Band Character:** Mainly built from **p-orbitals** which are directional $\rightarrow$ less overlap $\rightarrow$ less hopping $\rightarrow$ weaker coupling $\rightarrow$ flatter dispersion $\rightarrow$ **Heavier** m^* .
- **Conduction Band Character:** Mostly built from **s-orbitals** which are delocalized and isotropic $\rightarrow$ stronger overlap $\rightarrow$ stronger coupling $\rightarrow$ broader band $\rightarrow$ steeper dispersion $\rightarrow$ **Lighter** m^* .

[Pg 25]

5.3 The Hole

A hole is a **quasi-particle**.

- It implies the **absence of an electron** in a nearly full band.
- Its motion is determined by how surrounding electrons fill the empty state.
- It is not a "real" particle but a convenient way to describe collective behaviour.
- Holes generally have **lower mobility** than electrons.

[Pg 26]

Modification of Effective Mass Effective mass is modified in proximity to nuclei, i.e., the sources of periodic potential.

Stronger potential → **Stronger coupling**

↓

Near Brillouin zone edges, opens gaps and flattens bands.

Flatter bands → Heavier e^-

6 Group Velocity and Dispersion

6.1 Derivation

[Pg 27]

The group velocity is given by:

$$v_g = \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk}$$

(since $E = \hbar\omega$)

Recall for free electron:

$$E = \frac{\hbar^2 k^2}{2m} \implies \frac{dE}{dk} = \frac{\hbar^2 k}{m}$$

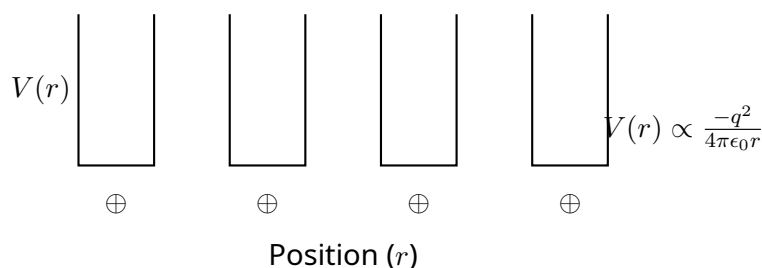
$$v_g = \frac{1}{\hbar} \left(\frac{\hbar^2 k}{m} \right) = \frac{\hbar k}{m} = \frac{p}{m} = v$$

Also, from dispersion relation:

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}$$

[Pg 28]

6.2 Periodic Potential Illustration



Band Extremes:

$$E_c = E_c + \frac{\hbar^2 k^2}{2m^*}$$

$$E_v = E_v - \frac{\hbar^2 k^2}{2m^*}$$

7 Brillouin Zones

7.1 Definition

[Pg 29]

Region in k-space closer to a given reciprocal lattice point than to any other.

1st Brillouin Zone: Set of all wavevectors k that can be reduced to the central cell without crossing a Bragg plane.

$$-\frac{\pi}{a} \leq k \leq \frac{\pi}{a} \quad (1\text{st BZ})$$

$$\frac{\pi}{a} \leq k \leq \frac{2\pi}{a} \quad (2\text{nd BZ})$$

[Pg 30]

Physics Inside 1st BZ Any physics in any band can be completely described using wave vectors inside the 1st Brillouin Zone because states outside can be reduced back using reciprocal lattice vectors. (Periodicity in k-space).

Shape of E-k Graph:

- Parabolic in **Low-k** limit.
- Curved in **High-k** limit (approaching BZ boundary).

[Pg 31]

Why Parabolic?

$$E(k) \approx E_c + \frac{\hbar^2 k^2}{2m^*}$$

Behaves almost like a free particle as $k \rightarrow 0$.

Note on Opto-electronics: When radiation falls on a semiconductor, it gets excited (super-excited) but relaxes back within femtoseconds.

At T=0:

- **Metals:** Have partially filled conduction bands $\rightarrow$ electrons can move freely.
- **Semiconductors:** Have empty C.B. at $T = 0$ but completely filled valence bands.

8 Density of States (DOS)

8.1 Fermi Level

[Pg 32]

This is the highest occupied electronic state (at T=0).

8.2 Derivation for 3-D Systems

Consider a system of length L (box). Spacing between each point in k -space is finite: $\frac{2\pi}{L}$.

By Bloch's theorem: $\Psi_k(x) = U_k(x)e^{ikx}$. Applying **Periodic Boundary Conditions**: $\Psi(x=0) = \Psi(x=L)$.

$$\Psi(x=0) = U_k(0)$$

$$\Psi(x=L) = U_k(x+L)e^{ikL} = U_k(x)e^{ikL}$$

$$\implies e^{ikL} = 1 \implies \cos(kL) + i \sin(kL) = 1$$

$$\implies kL = 2\pi n \implies k = \frac{2\pi n}{L}$$

[Pg 33]

Where $n = 1, 2, \dots$ Spacing: $\Delta k = \frac{2\pi}{L}$.

8.2.1 Number of States (1-D)

$$dN = \frac{dk}{2\pi/L} \times 2 = \frac{L}{\pi} dk$$

(The factor of 2 comes from spin of fermions).

$N_k \equiv$ Number density of state in k -space per unit length.

$$N_k = \frac{1}{\pi}$$

[Pg 34]

8.2.2 DOS in Energy Space

We need to find $D(E)$ such that $D(E)dE = N(k)dk$.

Using the parabolic approximation:

$$E = E_c + \frac{\hbar^2 k^2}{2m^*}$$

$$dE = \frac{\hbar^2 k}{m^*} dk \implies dk = \frac{m^*}{\hbar^2 k} dE$$

From 1D result:

$$dN = \frac{1}{\pi} dk = \frac{1}{\pi} \frac{m^*}{\hbar^2 k} dE$$

[Pg 35]

Substitute $k = \sqrt{\frac{2m^*(E-E_c)}{\hbar^2}}$:

$$dN = \frac{1}{\pi} \frac{m^*}{\hbar^2 \sqrt{\frac{2m^*(E-E_c)}{\hbar^2}}} dE$$

$$D(E) \propto \frac{1}{\sqrt{E-E_c}} \quad (1D \text{ Result})$$

[Pg 36]

8.2.3 3D Fermi Sphere Calculation

In 3D, $E = \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2 + k_z^2)$.

$$k = \sqrt{\frac{2m^*E}{\hbar^2}}$$

For a spherical shell of radius k and thickness dk , the volume of the shell is:

[Pg 37]

$$dV_k = 4\pi k^2 dk$$

Each allowed k-state occupies volume $(\frac{2\pi}{L})^3$. Number of states in shell (times 2 for spin):

$$dN = 2 \cdot \frac{4\pi k^2 dk}{(\frac{2\pi}{L})^3} = \frac{V}{\pi^2} k^2 dk$$

Substitute $k^2 = \frac{2m^*(E-E_c)}{\hbar^2}$ and $dk = \frac{m^*}{\hbar^2 k} dE$:

$$dN = \frac{V}{\pi^2} \left(\frac{2m^*(E-E_c)}{\hbar^2} \right) \frac{m^*}{\hbar^2 k} dE$$

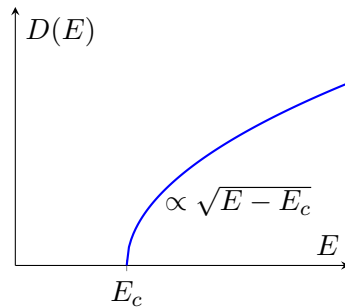
After simplifying:

[Pg 38]

$$dN = \frac{V}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} \sqrt{E-E_c} dE$$

Density of States Function:

$$D(E) = \frac{1}{V} \frac{dN}{dE} \propto \sqrt{E-E_c}$$



9 Fermi-Dirac Statistics

9.1 The Distribution Function

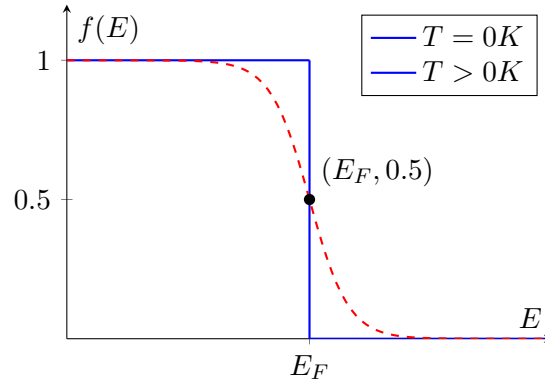
[Pg 39]

The probability of occupation of a state with energy E is given by the Fermi-Dirac distribution:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)}$$

$\uparrow$ Energy relative to Fermi Level
 $\downarrow$ Thermal Energy

Physical Interpretation: It relates how far the state is from the Fermi energy E_F .



10 Carrier Concentration (Intrinsic)

10.1 Equilibrium Condition

[Pg 40]

For an intrinsic semiconductor:

$$n = p$$

Using the approximations for the conduction and valence bands:

$$n = N_c e^{-(E_c - E_F)/k_B T}$$

$$p = N_v e^{-(E_F - E_v)/k_B T}$$

Equating them ($n = p$):

$$N_c e^{-(E_c - E_F)/k_B T} = N_v e^{-(E_F - E_v)/k_B T}$$

Taking natural logs:

$$\ln(N_c) - \frac{E_c - E_F}{k_B T} = \ln(N_v) - \frac{E_F - E_v}{k_B T}$$

Rearranging for E_F :

$$\frac{2E_F}{k_B T} = \frac{E_c + E_v}{k_B T} + \ln\left(\frac{N_v}{N_c}\right)$$

[Pg 41]

10.2 Intrinsic Fermi Level (E_i)

$$E_i = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln\left(\frac{N_v}{N_c}\right)$$

Approximation: Since effective masses are similar ($m_n^* \approx m_p^*$), $N_c \approx N_v$.

$$\ln(1) = 0 \implies E_i \approx \frac{E_c + E_v}{2} = \frac{E_G}{2}$$

The intrinsic Fermi level lies roughly in the middle of the band gap.

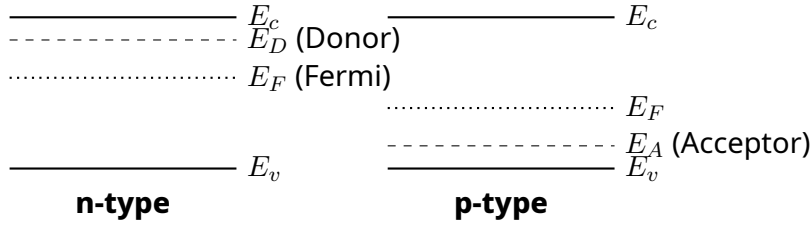
Data for Si ($T = 300K$):

- $E_g \approx 1.12 \text{ eV}$
- $n_i \approx 1.0 \times 10^{10} \text{ cm}^{-3}$ (Note: Notes say 10^{16} or similar, standard is 10^{10})

11 Extrinsic Semiconductors (Doping)

11.1 Types

1. **n-type:** Silicon doped with Pentavalent atoms (e.g., Phosphorus). Adds donor levels E_D .
2. **p-type:** Silicon doped with Trivalent atoms (e.g., Boron). Adds acceptor levels E_A .



where:

- N_D : Donor density
- N_A : Acceptor density

12 Derivation of Electron Density (n_0)

The electron concentration is the integral of the Density of States multiplied by the occupancy probability: [Pg 42]

$$n_0 = \int_{E_c}^{\infty} f(E)g(E) dE$$

Using the Boltzmann approximation ($E \gg E_F$, so $f(E) \approx e^{-(E-E_F)/kT}$) and 3D DOS:

$$n_0 = \int_{E_c}^{\infty} \frac{1}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c} \cdot e^{-(E-E_F)/k_B T} dE$$

Rearranging terms:

$$n_0 = \frac{1}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} e^{-(E_c-E_F)/k_B T} \int_{E_c}^{\infty} \sqrt{E - E_c} e^{-(E-E_c)/k_B T} dE$$

This integral is a standard form related to the Gamma function ($\Gamma(3/2) = \frac{\sqrt{\pi}}{2}$). [Pg 43]
Evaluating it yields the **Effective Density of States (N_c)**:

$$n_0 = N_c e^{-(E_c-E_F)/k_B T}$$

Where:

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2}$$

Similarly for holes:

$$p_0 = N_v e^{-(E_F-E_v)/k_B T}$$

Where $N_v = 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2}$.

12.1 Space Charge Neutrality

In equilibrium, the net charge density must be zero:

$$\rho = p_0 + N_D^+ - n_0 - N_A^- = 0$$

$$\implies p_0 - n_0 + N_D^+ - N_A^- = 0$$

Using the Law of Mass Action: $n_0 p_0 = n_i^2 \implies p_0 = n_i^2 / n_0$.

$$\frac{n_i^2}{n_0} - n_0 + N_D - N_A = 0$$

(Assuming complete ionization, $N_D^+ \approx N_D$).

13 Approximations

For n-type material ($N_D \gg N_A$ and $N_D \gg n_i$): We can approximate:

[Pg 44]

$$n_0 \approx N_D$$

Using the Mass Action Law for minority carriers (holes):

$$p_0 = \frac{n_i^2}{n_0} \approx \frac{n_i^2}{N_D}$$

Fermi Level Position (n-type) Since $n_0 = N_c e^{-(E_c - E_F)/kT} \approx N_D$:

$$E_c - E_F = k_B T \ln \left(\frac{N_c}{N_D} \right)$$

$$E_F = E_c - k_B T \ln \left(\frac{N_c}{N_D} \right)$$

Note: As doping (N_D) increases, the ratio $\frac{N_c}{N_D}$ decreases, $\ln(\dots)$ becomes smaller, and E_F moves closer to E_c . [Pg 45]

13.1 Most Probable Energy

We consider the carrier distribution function:

$$n(E) = f(E)g(E) \approx e^{-(E - E_F)/kT} \cdot C \sqrt{E - E_c}$$

(Let E be measured from E_c , so we look at term $\sqrt{E} e^{-E/kT}$).

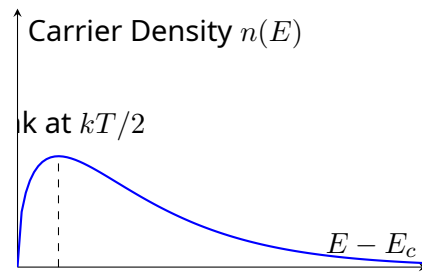
To find the peak (most probable energy for a carrier):

$$\frac{d}{dE} \left(E^{1/2} e^{-E/kT} \right) = 0$$

$$\frac{1}{2} E^{-1/2} e^{-E/kT} - \frac{1}{kT} E^{1/2} e^{-E/kT} = 0$$

$$\frac{1}{2E} - \frac{1}{kT} = 0 \implies E = \frac{kT}{2}$$

Thus, the peak of the electron distribution inside the band is at $\frac{k_B T}{2}$ above E_c .



14 Transport Mechanisms

14.1 Current Density Equations

[Pg 46]

Net Current = Drift + Diffusion.

1. Electron Current (J_n):

$$J_n(x) = \underbrace{qn\mu_n E_x}_{\text{Drift}} + \underbrace{qD_n \frac{dn}{dx}}_{\text{Diffusion}}$$

2. Hole Current (J_p):

$$J_p(x) = \underbrace{qp\mu_p E_x}_{\text{Drift}} - \underbrace{qD_p \frac{dp}{dx}}_{\text{Diffusion}}$$

- **Drift:** Motion due to Electric Field (μ = mobility).
- **Diffusion:** Motion due to concentration gradient (D = Diffusion coeff).

$$J_{total} = J_n + J_p$$

15 Continuity & Diffusion

15.1 In Absence of External E-Field

[Pg 56]

Consider only x-direction.

$$v_x = -\mu_x E_x \quad (\text{Drift})$$

Now consider the case of **Diffusion** (no net charge flow initially). Let τ be the average time between two successive collisions.

$N(t)$ = Number of electrons not having undergone a collision at time t . Rate of decrease of $N(t)$ is proportional to the number left unscattered:

$$-\frac{dN(t)}{dt} = \frac{N(t)}{\tau}$$

Solution:

$$N(t) = N_0 e^{-t/\tau}$$

[Pg 57]

15.2 Lecture 18: Flux and Concentration Gradient

Let $n(x)$ be the concentration of electrons. Consider a small element δx . **Flux** $\phi(x)$ is the number of particles crossing unit area per unit time.

Fick's Law of Diffusion:

$$\phi_n(x) = -D_n \frac{dn}{dx}$$

where D_n is the Diffusion Coefficient.

Similarly for holes:

$$\phi_p(x) = -D_p \frac{dp}{dx}$$

15.3 Diffusion Current Density

Since current $J = q \times \text{Flux}$:

$$J_{n,\text{diff}} = (-q)\phi_n(x) = (-q) \left(-D_n \frac{dn}{dx} \right) = +qD_n \frac{dn}{dx}$$

$$J_{p,\text{diff}} = (+q)\phi_p(x) = -qD_p \frac{dp}{dx}$$

16 Einstein Relation & Quasi-Fermi Levels

16.1 Equilibrium Check

[Pg 58]

Total Current $J = J_{\text{drift}} + J_{\text{diff}} = 0$ in equilibrium. For holes:

$$qp\mu_p E - qD_p \frac{dp}{dx} = 0$$

$$\mu_p p E = D_p \frac{dp}{dx}$$

Substituting $E = -\frac{dV}{dx}$ and $p = n_i e^{(E_i - E_F)/kT}$:

$$\mu_p p \left(-\frac{dV}{dx} \right) = D_p \frac{dp}{dx}$$

This leads to the **Einstein Relation**:

$$\frac{D}{\mu} = \frac{kT}{q}$$

16.2 Quasi-Fermi Levels

Under non-equilibrium (e.g., illumination or bias), $np \neq n_i^2$. We define:

$$n = n_i e^{(F_n - E_i)/kT}$$

$$p = n_i e^{(E_i - F_p)/kT}$$

Where F_n and F_p are **Quasi-Fermi levels**.

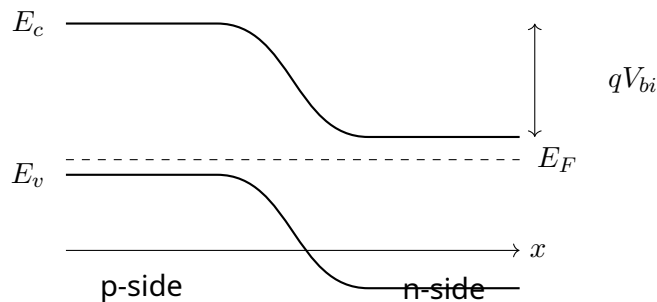
17 P-N Junction Physics

17.1 Band Diagram in Equilibrium

[Pg 59]

At equilibrium ($V = 0$):

- Fermi level E_F is constant throughout.
- Band bending occurs near the junction to align Fermi levels.



17.2 Built-in Potential (V_{bi} or V_0)

$$qV_{bi} = E_g - (E_c - E_F)_n - (E_F - E_v)_p$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

[Pg 60]

17.3 Current Components

$$J_n = \text{Drift} + \text{Diffusion}$$

Using Quasi-Fermi levels, the total current can be expressed compactly as:

$$J_n = n\mu_n \frac{dF_n}{dx}$$

$$J_p = p\mu_p \frac{dF_p}{dx}$$

18 Lecture 19: The Diode Equation

18.1 Biasing

[Pg 62]

1. **Forward Bias (V_f):** P-side is positive. Barrier height decreases by qV_f . Diffusion current increases exponentially.
2. **Reverse Bias (V_r):** N-side is positive. Barrier height increases by qV_r . Current is very small (leakage).

18.2 Shockley Diode Equation

The net current I is given by:

$$I = I_0 \left(e^{qV/kT} - 1 \right)$$

Where I_0 is the reverse saturation current:

$$I_0 = qA \left(\frac{D_p}{L_p} p_{n0} + \frac{D_n}{L_n} n_{p0} \right)$$

- L_p, L_n : Diffusion lengths ($L = \sqrt{D\tau}$).
- p_{n0}, n_{p0} : Equilibrium minority carrier concentrations.

[Pg 63]

18.3 Depletion Width (W)

In equilibrium:

$$W = \sqrt{\frac{2\epsilon V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Under reverse bias V_R , V_{bi} is replaced by $V_{bi} + V_R$, so W increases.

19 Rectifier Circuits

19.1 Introduction

[Pg 73]

A P-N junction diode acts as a rectifier (converts AC to pulsating DC).

[Pg 74]

19.2 Half-Wave Rectifier (HWR)

A single diode connected in series with a load R_L .

- Conducts only during positive half-cycle.
- **Average DC Voltage:** $V_{dc} = \frac{V_m}{\pi}$
- **RMS Voltage:** $V_{rms} = \frac{V_m}{2}$
- **Efficiency:** $\eta = \frac{P_{dc}}{P_{ac}} \approx 40.6\%$

[Pg 75]

19.3 Full-Wave Rectifier (FWR)

Uses two diodes (center-tapped transformer).

- Conducts during both half-cycles.
- **Average DC Voltage:** $V_{dc} = \frac{2V_m}{\pi}$
- **RMS Voltage:** $V_{rms} = \frac{V_m}{\sqrt{2}}$
- **Efficiency:** $\eta \approx 81.2\%$

[Pg 77]

19.4 Bridge Rectifier

Uses 4 diodes. No center tap needed.

- Same efficiency as FWR (81.2%).
- Peak Inverse Voltage (PIV) is V_m (unlike $2V_m$ for center-tapped FWR).

[Pg 79]

19.5 Ripple Factor (γ)

Measure of AC component in output.

$$\gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1}$$

- HWR: $\gamma = \sqrt{\frac{\pi^2}{4} - 1} \approx 1.21$
- FWR: $\gamma = \sqrt{\frac{\pi^2}{8} - 1} \approx 0.48$

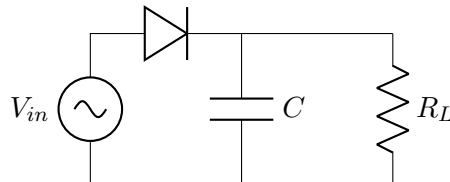
20 Filters

20.1 Capacitor Filter

[Pg 80]

To smoothen the pulsating DC, a capacitor is connected in parallel with R_L .

- **Charging:** When diode is forward biased (fast, time constant $\tau_c = R_f C$).
- **Discharging:** When diode is reverse biased (slow, time constant $\tau_d = R_L C$).



[Pg 81]

20.2 Ripple Calculation

Approximation for small ripple ($R_L C$ is large):

$$V_{dc} \approx V_m - \frac{V_r}{2}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC}$$

Ripple Factor with Filter (FWR):

$$\gamma = \frac{1}{4\sqrt{3}fR_L C}$$

[Pg 85]

20.3 Inductor Filter

An inductor in series opposes change in current. It is better suited for heavy loads (low R_L) where current is high.

21 Diode As a Clipper

21.1 Introduction

[Pg 86]

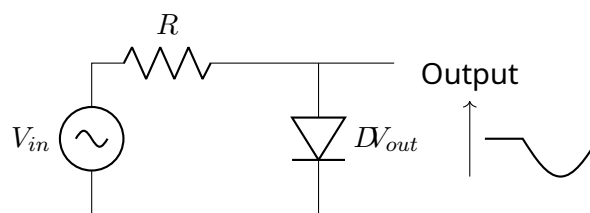
Clippers are circuits that remove (clip) a portion of an input signal without distorting the remaining part of the waveform.

21.2 Types of Clippers

21.2.1 1. Positive Clipper

Removes the positive half cycles of the input signal.

- **During +ve cycle:** Diode is Forward Biased (ON) → Acts as a short circuit (if ideal). $V_{out} = 0$.
- **During -ve cycle:** Diode is Reverse Biased (OFF) → Open circuit. $V_{out} = V_{in}$.



21.2.2 2. Negative Clipper

Removes the negative half cycles.

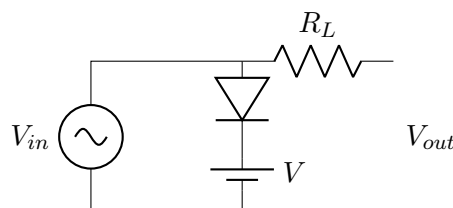
- Diode direction is reversed compared to positive clipper.
- **During -ve cycle:** Diode conducts (ON) → $V_{out} = 0$.

[Pg 87]

21.3 Biased Clippers

We can add a DC battery (V) to shift the clipping level.

Example:

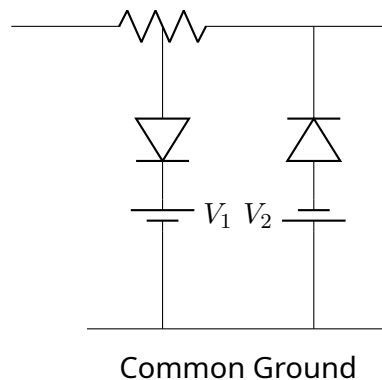


- For $V_{in} < V$: Diode is OFF.
- For $V_{in} > V$: Diode conducts.

[Pg 88]

21.4 Combination Clipper

Can clip both positive and negative peaks.



Output is a "sliced" waveform between V_1 and $-V_2$.

22 Clamping Circuits

22.1 Definition

[Pg 89]

1. A Clamping circuit places the positive or negative peaks of a signal at a desired DC level.
2. Essentially adds a DC shift to the AC signal without changing its shape (Peak-to-Peak value remains same).

22.2 Circuit Structure

Uses a Capacitor (C), Diode (D), and Resistor (R).

$$\tau = RC$$

Condition: The time constant τ must be large enough ($\tau \gg T$) so the capacitor doesn't discharge significantly during the non-conducting period.

[Pg 90]

22.3 Positive Clamper

Shifts the waveform **up**.

- During -ve half cycle, Diode is ON $\rightarrow$ Capacitor charges to V_m (with polarity + on right).
- During +ve half cycle, Diode is OFF $\rightarrow$ Capacitor acts as a battery of voltage V_m .
- Total Output: $V_{out} = V_{in} + V_m$.

Peaks:

[Pg 91]

- Original: $+V_m$ to $-V_m$.
- Clamped: $+2V_m$ to 0.

23 Zener Diode

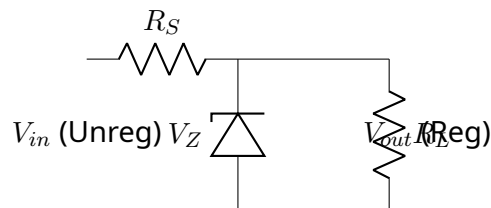
23.1 Characteristics

[Pg 92]

Designed to operate in the **Reverse Breakdown** region.

- V_z : Zener Breakdown Voltage.
- Can be used as a **Voltage Regulator**.

23.2 Voltage Regulator Circuit



Current Equation:

$$I_S = I_Z + I_L$$
$$I_S = \frac{V_{in} - V_Z}{R_S}$$

[Pg 93]

23.3 Regulation Types

1. **Line Regulation:** V_{in} varies, R_L is fixed. V_{out} remains constant (V_Z) as long as I_Z stays within limits ($I_{Zmin} < I_Z < I_{Zmax}$).
2. **Load Regulation:** V_{in} is fixed, R_L varies. V_{out} remains constant.

24 Bipolar Junction Transistor (BJT)

24.1 Structure

[Pg 94]

Three-terminal device with 3 layers.

- **PNP** or **NPN**.
- Terminals: Emitter (E), Base (B), Collector (C).

Two Junctions: 1. Emitter-Base Junction (JE). 2. Collector-Base Junction (JC).

[Pg 95]

24.2 Doping and Size Profile

- **Emitter:** Heavily doped (Injects carriers). Moderate size.
- **Base:** Lightly doped (Reduces recombination). Very thin.
- **Collector:** Moderately doped. Largest size (Dissipates heat).

24.3 Operation Mode (Active)

For amplification:

- **E-B Junction:** Forward Biased.
- **C-B Junction:** Reverse Biased.

[Pg 97]

24.4 Current Equation

$$I_E = I_B + I_C$$

Since base is thin and lightly doped, recombination is low ($\approx 5\%$), so:

$$I_C \approx I_E$$

$$I_B \ll I_E$$

[Pg 99]

24.5 Numbering Scheme (Pro-tip)

Standard naming:

- First letter: Material (A = Ge, B = Si).
- Second letter: Function (C = Audio/LF, F = HF).
- Example: BC107 (Silicon, Audio Freq).

25 Transistor Configurations

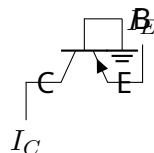
Three modes based on the common terminal:

[Pg 100]

1. Common Base (CB)
2. Common Emitter (CE)
3. Common Collector (CC)

25.1 Common Base (CB) Configuration

[Pg 101]



Used as High Frequency amplifier.

Current Amplification Factor (α):

$$\alpha = \frac{\text{Output Current}}{\text{Input Current}} = \frac{I_C}{I_E}$$

Since $I_C < I_E$, $\alpha < 1$ (typically 0.9 to 0.99).

[Pg 102]

25.1.1 Collector Current Expression

I_C consists of: 1. Portion of emitter current reaching collector (αI_E). 2. Leakage current due to reverse biased BC junction (I_{leak} or I_{CBO}).

$$I_C = \alpha I_E + I_{CBO}$$

[Pg 104]

25.1.2 Input Characteristics (CB)

Plot of I_E vs V_{EB} at constant V_{CB} .

- Resembles a forward-biased diode characteristic.
- I_E increases rapidly for small increase in V_{EB} .
- **Input Resistance (r_i):** Very low (few Ω).

[Pg 105]

25.1.3 Output Characteristics (CB)

Plot of I_C vs V_{CB} at constant I_E .

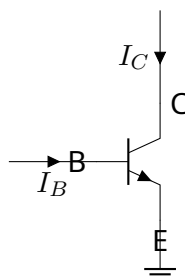
- **Active Region:** I_C depends only on I_E , independent of V_{CB} . Curves are flat.
- **Saturation Region:** $V_{CB} < 0$ (Forward biased). Current collapses.
- **Output Resistance (r_o):** Very high ($\sim M\Omega$) due to flat curves.

26 Common Emitter (C.E.) Configuration

This is the most important configuration for amplifiers.

[Pg 106]

- **Emitter** is common to both input and output.
- **Input:** Applied between Base and Emitter.
- **Output:** Taken between Collector and Emitter.



26.1 Current Amplification Factor (β)

Defined as the ratio of output current to input current:

$$\beta = \frac{I_C}{I_B}$$

Typically β ranges from 20 to 500.

[Pg 107]

26.2 Relation between β and α

We know $I_E = I_B + I_C$. Dividing by I_C :

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1 \implies \frac{1}{\beta} = \frac{1}{\alpha} - 1 = \frac{1 - \alpha}{\alpha}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

As $\alpha \rightarrow 1$, $\beta \rightarrow \infty$.

26.3 Collector Current & Leakage (I_{CEO})

For Common Base: $I_C = \alpha I_E + I_{CBO}$. Substituting $I_E = I_C + I_B$:

$$I_C = \alpha(I_C + I_B) + I_{CBO}$$

$$I_C(1 - \alpha) = \alpha I_B + I_{CBO}$$

$$I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

We define the **Collector-Emitter Leakage Current** (I_{CEO} , with Base Open):

$$I_{CEO} = \frac{1}{1 - \alpha} I_{CBO} = (1 + \beta) I_{CBO}$$

Total Current Equation:

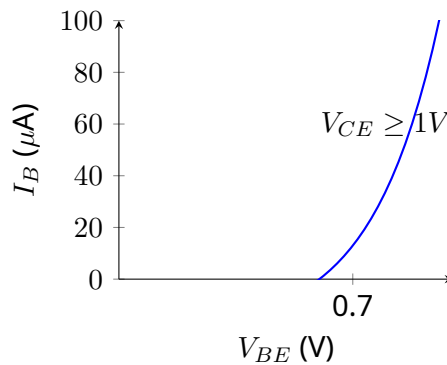
$$I_C = \beta I_B + I_{CEO}$$

Note: I_{CEO} is much larger than I_{CBO} .

[Pg 108]

26.4 Input Characteristics (CE)

Plot of I_B vs V_{BE} at constant V_{CE} .

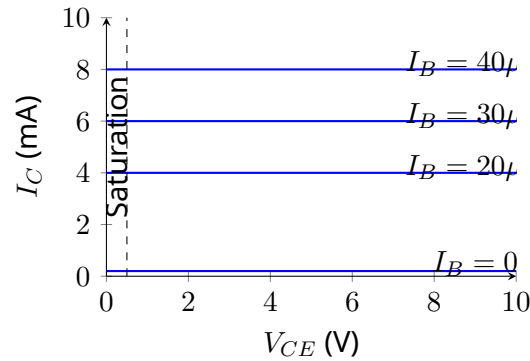


- Resembles a forward-biased diode.
- Input resistance r_i is higher than CB configuration but still low ($< 1k\Omega$).

[Pg 109]

26.5 Output Characteristics (CE)

Plot of I_C vs V_{CE} at constant I_B .



Regions of Operation:

1. **Active Region:** $I_C = \beta I_B$. Used for amplification.
2. **Saturation Region:** $V_{CE} < V_{CE(sat)} \approx 0.2V$. Current is limited by external circuit, not β .
3. **Cutoff Region:** $I_B = 0$. Only leakage current I_{CEO} flows.

27 Common Collector (C.C.)

Also known as **Emitter Follower**.

[Pg 111]

- **Input:** Base-Collector (Collector is AC ground).
- **Output:** Emitter-Collector.

Current Gain (γ):

$$\gamma = \frac{I_E}{I_B} = \frac{I_C + I_B}{I_B} = \beta + 1$$

$$\gamma = \frac{1}{1 - \alpha}$$

Comparison Table:

[Pg 112]

Characteristic	CB	CE	CC
Input Resistance	Low ($\sim 100\Omega$)	Moderate ($\sim 750\Omega$)	Very High ($\sim 750k\Omega$)
Output Resistance	Very High ($\sim 450k\Omega$)	High ($\sim 45k\Omega$)	Low ($\sim 50\Omega$)
Voltage Gain	High (~ 150)	High (~ 500)	Low (≈ 1)
Current Gain	< 1 (α)	High (β)	High (γ)
Application	HF Apps	Audio Amp	Buffer / Matching

28 CE Amplifier & Analysis

28.1 Circuit Diagram

[Pg 113]

To work as an amplifier, we apply: 1. DC Bias (V_{BB}, V_{CC}) to set the operating point (Q-point). 2. AC Input Signal.

Biassing Rule:

- E-B Junction: Forward Biased.
- C-B Junction: Reverse Biased.

[Pg 114]

28.2 DC Load Line Analysis

Applying KVL to the collector loop:

$$V_{CC} - I_C R_L - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_L$$

This is a linear equation ($y = mx + c$) plotting I_C vs V_{CE} .

- **Y-intercept (Saturation):** $V_{CE} = 0 \implies I_C = V_{CC}/R_L$.
- **X-intercept (Cutoff):** $I_C = 0 \implies V_{CE} = V_{CC}$.

[Pg 115]

28.3 Operating Point (Q-Point)

The intersection of the DC Load Line and the transistor's output characteristic curve for a specific base current I_B . * **Quiescent:** Means "silent". It is the DC values (I_{CQ}, V_{CEQ}) when no AC signal is applied.

[Pg 116]

28.4 Cutoff and Saturation Points

- **Cutoff:** Point B. $I_B = 0$. Collector current is approx zero (I_{CEO}). $V_{CE} \approx V_{CC}$. Switch is OPEN.
- **Saturation:** Point A. Maximum current flows. $V_{CE} \approx 0.2V$ (Knee voltage). Switch is CLOSED.

29 Transistor as a Switch

BJT operates in two extreme regions: Cutoff (OFF) and Saturation (ON).

[Pg 117]

29.1 NPN Switch

Emitter grounded.

Base Input	State	Region	Output (V_{CE})
LOW (0V)	OFF	Cutoff	HIGH (V_{CC})
HIGH ($> 0.7V$)	ON	Saturation	LOW ($\approx 0.2V$)

Logic: Inverter (NOT Gate).

[Pg 118]

29.2 PNP Switch

Emitter connected to $+V_{CC}$.

- **Input Low:** $V_B < V_E$. E-B is Forward Biased $\rightarrow$ ON. Output High.
- **Input High:** $V_B \approx V_{CC}$. E-B is not Forward Biased $\rightarrow$ OFF. Output Low (pulled down).

30 Limitations of BJT

In a BJT, current is largely due to **minority carrier diffusion** across the base.

[Pg 119]

- Diffusion is a relatively slow process compared to drift.
- **Base Charging:** At high frequencies (GHz), the carriers cannot diffuse fast enough to track the signal.
- **Storage Time:** When switching off, minority carriers stored in the base must be removed, causing delay.

Solution: Field Effect Transistors (FET).

31 Field Effect Transistor (FET)

31.1 Key Differences from BJT

[Pg 120]

1. **Transport:** Primary cause of charge transport is **Drift** (in an electric field). BJT is Diffusion-based.
2. **Carriers:** FET is a **Unipolar** device (only majority carriers take part: either electrons or holes). BJT is Bipolar (e^- and h^+).
3. **Control:** FET is **Voltage Controlled** (Voltage at Gate controls current). BJT is Current Controlled (I_B controls I_C).

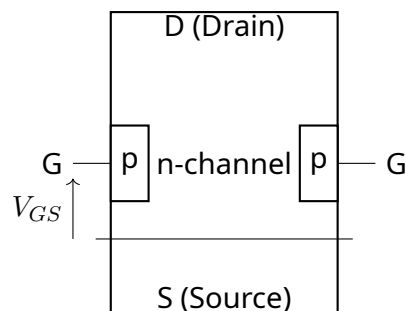
32 Field Effect Transistor (FET)

32.1 Channel Type

[Pg 121]

The type of FET depends on its construction and doping.

1. **Type of carrier conducts:** Either n-type (electrons) or p-type (holes).
2. **n-channel FET:** $p \gg n$ (Gate is heavily doped p-type, Channel is n-type).
3. **p-channel FET:** Channel is p-type.



Terminals:

- **G (Gate):** Heavily doped. Controls channel conductivity.
- **S (Source):** Where majority carriers enter.
- **D (Drain):** Where majority carriers leave.
- **Channel:** Region between Source and Drain made of n/p-type semiconductor.

[Pg 122]

32.2 Biasing of FET

- **Gate-Source (G-S) Junction:** Always kept at **Reverse Bias**.
- **Input Impedance:** Since the G-S junction is reverse biased, $I_G \approx 0$. Thus, FET has a much higher input impedance than BJT.
- **Drain Current:** Flows from Drain to Source (for n-channel electrons flow $S \rightarrow D$).
- **Current Equality:** Source current I_S = Drain current I_D .

Working Principle:

- G-S is in Reverse Bias (R.B.).
- G-S depletion region width depends upon V_{GS} .
- ****Higher V_{GS} (more negative for n-channel) $\rightarrow$ Narrower the channel $\rightarrow$ Lesser the I_{SD} .****
- At $V_{GS} = 0$, the channel has the **maximum width**.

33 Depletion Region & Pinch-Off

33.1 Effect of Drain Voltage (V_{DS})

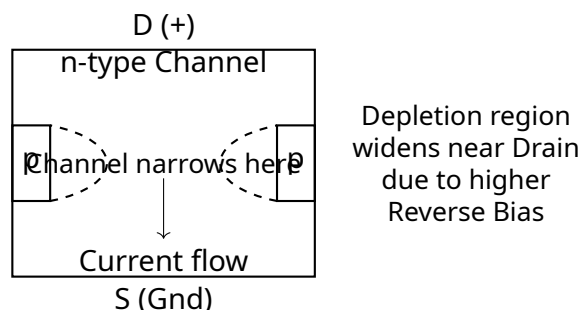
[Pg 123]

As V_{DS} increases (with V_{GS} fixed, say at 0V):

1. The voltage drop across the n-type channel increases linearly from Source to Drain.
2. ****Voltage at the Drain end rises relative to the Source.****
3. Consequently, the Gate-Channel junction near the Drain is **more reverse-biased** than near the Source.
4. **Result:** The depletion region near the Drain widens more than near the Source.

[Pg 124]

33.2 Diagrammatic Representation



- As V_{DS} increases, the depletion regions expand.
- Eventually, they effectively "touch" or "pinch" the channel near the drain end.
- This state is called **Pinch-off**.

Note: The channel never completely closes; a small gap remains to allow the saturation current to flow. If it closed completely, current would stop, removing the voltage drop that caused the pinch-off (a contradiction).

34 Drain Characteristics (I_D vs V_{DS})

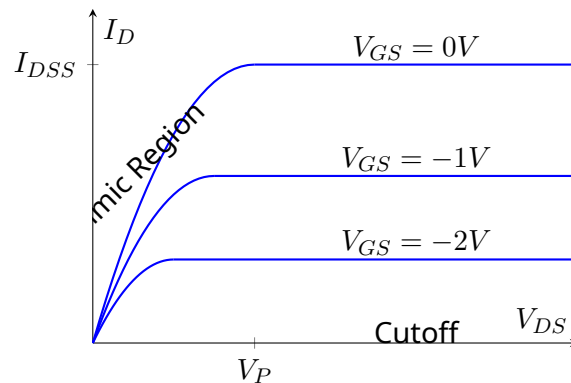
The curve relates Drain Current I_D to Drain-Source Voltage V_{DS} for various fixed Gate-Source Voltages V_{GS} . [Pg 125]

1. **Ohmic Region:** For small V_{DS} , I_D rises linearly with V_{DS} . The FET behaves like a voltage-controlled resistor.
2. **Pinch-off Voltage (V_P):** The value of V_{DS} at which the current saturates.
3. **Saturation (Active) Region:** Above V_P , the current I_D remains constant despite increasing V_{DS} .

4. Equation:

$$V_{DS} = V_P \Rightarrow I_D = I_{DSS}$$

where I_{DSS} is the saturation current at $V_{GS} = 0$.



35 FET Transfer Characteristics

35.1 Shockley's Equation

[Pg 126]

While the Drain Characteristic plots I_D vs V_{DS} , the **Transfer Characteristic** plots I_D vs V_{GS} (Input voltage controlling Output current).

In the **Saturation (Active) Region**, the relationship is given by:

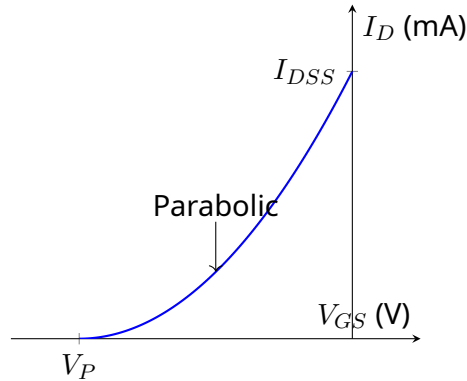
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

- I_{DSS} : Drain-to-Source Saturation Current (Maximum current when $V_{GS} = 0$).
- V_P : Pinch-off Voltage (Voltage at which I_D becomes zero).

[Pg 127]

35.2 Transfer Curve Plot

This is a parabolic curve.



Key Points on Curve:

1. At $V_{GS} = 0 \Rightarrow I_D = I_{DSS}$.
2. At $V_{GS} = V_P \Rightarrow I_D = 0$.
3. At $V_{GS} = V_P/2 \Rightarrow I_D = I_{DSS}/4$.

36 Transconductance (g_m)

36.1 Definition

[Pg 128]

Transconductance represents the gain of the FET. It defines how much the drain current changes for a small change in gate-source voltage.

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS}=\text{const}} = \frac{\partial I_D}{\partial V_{GS}}$$

36.2 Derivation

Differentiating Shockley's Equation w.r.t V_{GS} :

$$g_m = -\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right)$$

Since V_P is negative for n-channel, let $g_{m0} = -\frac{2I_{DSS}}{V_P}$ (Maximum transconductance at $V_{GS} = 0$).

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_P} \right)$$

37 Ebers-Moll Model

37.1 Introduction

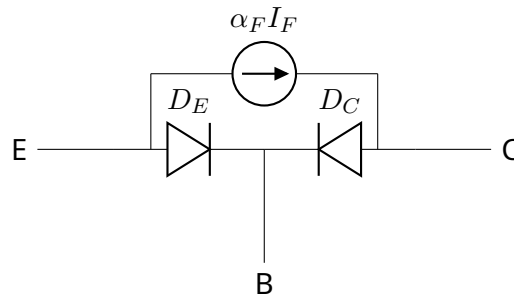
[Pg 129]

The Ebers-Moll model is a comprehensive **Large Signal Model** for the Bipolar Junction Transistor (BJT). It describes the transistor's behavior in **all regions** (Active, Saturation, Cutoff) by relating terminal currents to junction voltages.

[Pg 130]

37.2 Two-Diode Analogy

The BJT is modeled as two diodes connected back-to-back, interacting via current-controlled current sources.



[Pg 131]

37.3 Model Equations

$$I_E = I_{ES}(e^{V_{BE}/V_T} - 1) - \alpha_R I_{CS}(e^{V_{BC}/V_T} - 1)$$

$$I_C = \alpha_F I_{ES}(e^{V_{BE}/V_T} - 1) - I_{CS}(e^{V_{BC}/V_T} - 1)$$

****Reciprocity Relation:**** $\alpha_F I_{ES} = \alpha_R I_{CS}$.

38 BJT Biasing

38.1 Need for Biasing

[Pg 132]

Biasing applies DC voltages and currents to: 1. Set up a stable ****Operating Point (Q-point)****. 2. Ensure the device stays in the ****Active Region****. 3. Prevent drift into Saturation/Cutoff due to temperature.

[Pg 133]

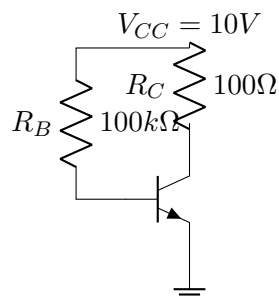
38.2 Current Source Representation

In active mode, the Collector-Base junction behaves like a ****Current Dependent Current Source****: $I_C = \beta I_B$.

[Pg 134]

38.3 Example 1: Fixed Bias Configuration

(n-p-n Configuration Only)



[Pg 135]

38.4 Analysis

Applying KVL to Base Loop:

$$V_{CC} - I_B R_B - V_{BE} = 0 \implies I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Given values ($V_{CC} = 10V$, $R_B = 100k\Omega$, $\beta = 100$):

$$I_B = \frac{10 - 0.7}{100k} = 93\mu A$$

$$I_C = \beta I_B = 9.3mA$$

$$V_{CE} = V_{CC} - I_C R_C = 10 - (9.3m)(100) = 9.07V$$

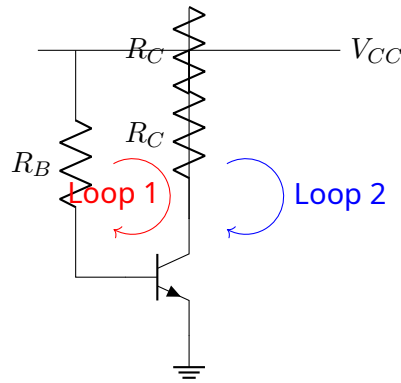
39 Biasing Examples Continued

39.1 Example 2: Fixed Bias with Common Supply

[Pg 136]

(n-p-n Configuration)

In practice, the base and collector resistors are connected to a single DC supply V_{CC} .



Loop 1 (Base-Emitter):

$$V_{CC} - I_B R_B - V_{BE} = 0$$

Since $V_{BE} \approx 0.7V$ (or V_{Th}), we get:

$$I_B = \frac{V_{CC} - V_{Th}}{R_B}$$

Loop 2 (Collector-Emitter):

[Pg 137]

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

Since $I_C = \beta I_B$, we can substitute:

$$V_{CE} = V_{CC} - \beta I_B R_C$$

$$V_{CE} = V_{CC} - \beta \left(\frac{V_{CC} - V_{Th}}{R_B} \right) R_C$$

$$V_{CE} = V_{CC} \left(1 - \beta \frac{R_C}{R_B} \right) \quad (\text{Assuming } V_{CC} \gg V_{Th})$$

Condition for Active Region ($V_{CE} > 0$):

$$1 - \beta \frac{R_C}{R_B} > 0 \implies \frac{R_B}{R_C} > \beta$$

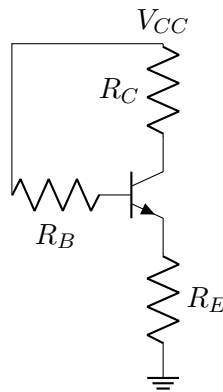
39.2 Example 3: Circuit Analysis

Given:

[Pg 138]

- $R_B = 430k\Omega$
- $R_C = 2k\Omega$
- $R_E = 1k\Omega$
- $V_{CC} = 20V, \beta = 50$

Find: $I_B, I_C, V_{CE}, V_C, V_E, V_B, V_{BC}$.



Solution:

1. Base Loop:

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

Substitute $I_E = (1 + \beta)I_B$:

$$V_{CC} - V_{BE} = I_B [R_B + (1 + \beta)R_E]$$

$$I_B = \frac{20 - 0.7}{430k + (51)(1k)} = \frac{19.3}{481k} \approx 40.1\mu A$$

2. Collector Current:

$$I_C = \beta I_B = 50 \times 40.1\mu A \approx 2.005mA$$

3. Emitter Current:

$$I_E \approx I_C = 2.005mA$$

4. Voltages:

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$V_{CE} = 20 - (2.005)(2) - (2.005)(1) \approx 13.99V$$

40 Small Signal Analysis

To analyze the transistor as an amplifier (AC analysis), we use ****Two-Port Networks****. [Pg 139]

40.1 Two-Port Network



Why not use Z-parameters ($V = ZI$)? * In Z-parameters, every element has dimensions of resistance (Ω). * Transistors have mixed properties (current gains, conductance, etc.). * Hence, we use **h-parameters (Hybrid Parameters)**.

[Pg 140]

40.2 h-Parameter Equations

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

Parameter Definitions:

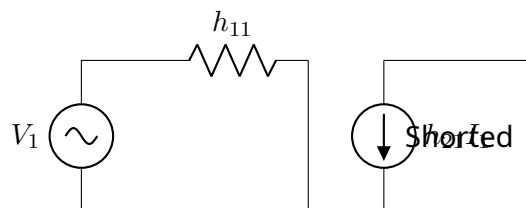
- $h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$ (Input Impedance, Ω)
- $h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$ (Reverse Voltage Ratio, dimensionless)
- $h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$ (Forward Current Gain, dimensionless)
- $h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$ (Output Admittance, Siemens/mho)

41 Hybrid Model Analysis

41.1 Forward Characteristics (Output Shorted)

[Pg 141]

Set $V_2 = 0$ (Output Shorted).

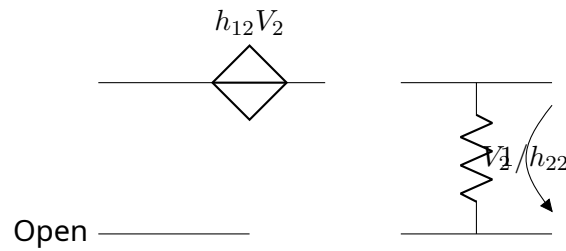


* Input Impedance $h_{11} = V_1/I_1$. * Forward Current Gain $h_{21} = I_2/I_1$.

[Pg 142]

41.2 Reverse Characteristics (Input Open)

Set $I_1 = 0$ (Input Open).

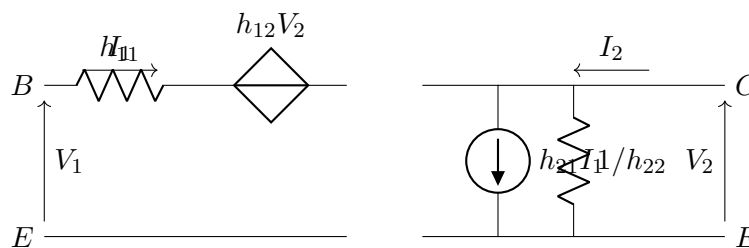


* Reverse Voltage Gain $h_{12} = V_1/V_2$. * Output Admittance $h_{22} = I_2/V_2$.

[Pg 143]

41.3 Equivalent Circuit

Combining both parts, we get the full Hybrid Model:



Equations:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

42 Transistor h-Parameters

The parameters are renamed based on the configuration (CB, CE, CC). * First subscript: function (i =input, r =reverse, f =forward, o =output). * Second subscript: configuration (b, e, c).

[Pg 144]

For Common Base (CB):

- $h_{ib} = h_{11}$ (Input Impedance) $\approx r_e$ (Dynamic resistance of diode).
- $h_{fb} = h_{21}$ (Forward Current Gain) $\approx -\alpha \approx -1$.
- $h_{rb} = h_{12}$ (Reverse Voltage Ratio) ≈ 0 .
- $h_{ob} = h_{22}$ (Output Admittance) ≈ 0 (Ideal current source).

For Common Emitter (CE):

[Pg 145]

- $h_{ie} = h_{11} \approx \beta r_e$.
- $h_{fe} = h_{21} = \beta$.
- $h_{re} = h_{12}$ (Small value).
- $h_{oe} = h_{22}$ (Small admittance).

[Pg 146]

42.1 Table of h-Parameters

Parameter	Common Base (CB)	Common Emitter (CE)
Input Imp. (h_i)	h_{ib}	h_{ie}
Fwd Gain (h_f)	h_{fb}	h_{fe}
Rev Gain (h_r)	h_{rb}	h_{re}
Output Adm. (h_o)	h_{ob}	h_{oe}